

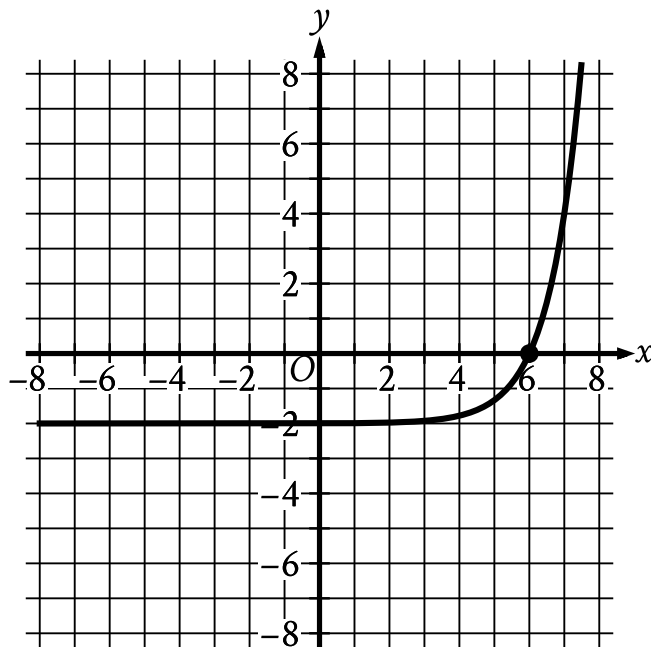
● EASY

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

02



- Moving from left to right:
 - The curve passes from quadrant 3 to quadrant 4 to quadrant 1.
 - In quadrant 3, the curve trends up gradually to the y axis.
 - In quadrant 4, the curve trends up gradually to the point (6 comma 0).
 - In quadrant 1, the curve trends up sharply.
- As x decreases, the curve approaches the line y equals negative 2.
- The curve passes through the following points:
 - (5 comma negative four thirds)
 - (6 comma 0)

What is the x -coordinate of the x -intercept of the graph shown?

STUDENT-PRODUCED RESPONSE

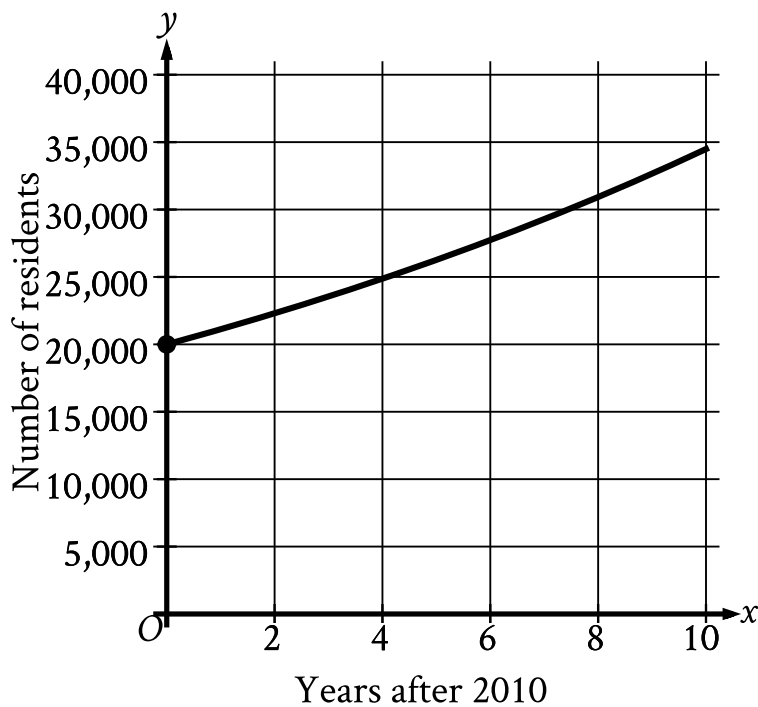
.	/	.	.
(	)	(	)

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function f is defined by $fx = 8\sqrt{x}$. For what value of x does $fx = 48$?

- A. 6
- B. 8
- C. 36
- D. 64

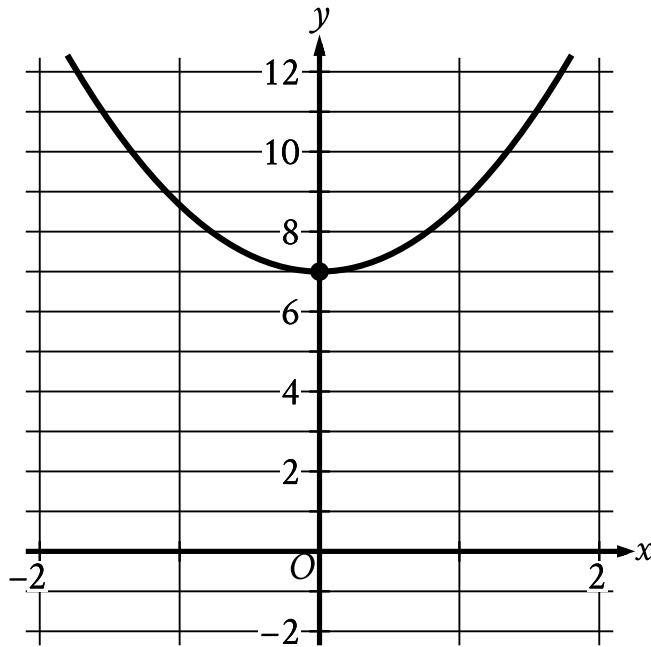


- The curve is in quadrant 1.
- The curve trends gradually up from left to right.
- The curve begins at the point (0 comma 20,000).
- The curve passes through the following points:
 - (0 comma 20,000)
 - approximately (5 comma 26,263)
 - approximately (10 comma 34,488)

The graph shown models the number of residents of a certain city x years after 2010. How many residents does this model estimate the city had in 2010?

- A. 0
- B. 2,000
- C. 20,000

D. 25,000



- The parabola opens upward.
- The vertex is at the point (0 comma 7).
- The parabola passes through the following points:
 - (negative 1 comma StartFraction 26 Over 3 EndFraction)
 - (0 comma 7)
 - (1 comma StartFraction 26 Over 3 EndFraction)

The parabola shown intersects the y -axis at the point x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function k is defined by $kx = x^3 + 110$. What is the value of kx when $x = 2$?

A. 118

B. 116

C. 115

D. 110

$$P(t) = 1,800(1.02)^t$$

The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. What is the best interpretation of $P(0) = 1,800$ in this context?

- A. The estimated number of marine mammals in the area was 102 when the study began.
- B. The estimated number of marine mammals in the area was 1,800 when the study began.
- C. The estimated number of marine mammals in the area increased by 102 each year during the study.
- D. The estimated number of marine mammals in the area increased by 1,800 each year during the study.

The function g is defined by $gx = \sqrt{8x + 1}$. What is the value of $g3$?

A. $\frac{5}{8}$

B. $\frac{25}{8}$

C. 5

D. 25

The function f is defined by $fx = \frac{16}{x}$. What is the value of fx when $x = 17$?

A. $\frac{16}{17}$

B. $\frac{17}{16}$

C. 16

D. 17

The function g is defined by $gx = \sqrt{x} + 300$. What is the value of gx when $x = 81$?

- A. 9
- B. 300
- C. 309
- D. 381

The function g is defined by $gx = x^2 + 9$. For which value of x is $gx = 25$?

- A. 4
- B. 5
- C. 9
- D. 13